

Prompt Engineering Showcase

Task 1 - VedGrow Generative AI Internship

Objective

This document demonstrates 25 effective prompts across five domains: coding, writing, summarization, analysis, and creative work. Each example includes a bad prompt, an optimized prompt, and the reasoning behind the improvement.

Prompt Types Covered

- **Zero-shot prompting:** Asking the model to perform a task without examples.
- **Few-shot prompting:** Providing examples so the model can follow the desired pattern.
- **Reasoning-summary prompting:** Asking the model to solve step by step internally and provide a concise explanation or key steps, rather than asking for hidden chain-of-thought.

Domains Covered

1. Coding
2. Writing
3. Summarization
4. Analysis
5. Creative

Prompt 1: Python Bug Fixing

Domain: Coding

Prompt Type: Zero-shot

Use Case: Debugging Python code with explanation.

Bad Prompt

Fix this code.

Optimized Prompt

You are a Python debugging assistant. Review the code below, identify the error, explain the cause briefly, and provide a corrected version. Also mention one best practice to avoid this issue in the future.

Code:

paste code here

Why This Is Better

Adds role, context, specific output structure, and a learning goal.

Prompt 2: SQL Query Generation

Domain: Coding

Prompt Type: Few-shot

Use Case: Generating SQL queries from natural language.

Bad Prompt

Write SQL for sales.

Optimized Prompt

Convert the user request into a SQL query. Follow the examples.

Example 1:

Request: Show total sales by month.

SQL: SELECT DATE_TRUNC('month', order_date) AS month, SUM(amount) AS total_sales FROM orders GROUP BY month ORDER BY month;

Example 2:

Request: Show the top 5 customers by revenue.

SQL: SELECT customer_id, SUM(amount) AS revenue FROM orders GROUP BY customer_id ORDER BY revenue DESC LIMIT 5;

Now write SQL for this request:

Request: Show average order value by product category for 2025.

Why This Is Better

Provides examples, expected style, and schema assumptions, improving accuracy.

Prompt 3: Algorithm Explanation

Domain: Coding

Prompt Type: Reasoning summary

Use Case: Learning an algorithm.

Bad Prompt

Explain binary search.

Optimized Prompt

Explain binary search for a beginner. Include: purpose, step-by-step process, Python implementation, time complexity, and one simple example trace. Keep the explanation clear and concise.

Why This Is Better

Specifies audience, structure, code language, and required complexity details.

Prompt 4: Code Review

Domain: Coding

Prompt Type: Zero-shot

Use Case: Improving code quality.

Bad Prompt

Review my code.

Optimized Prompt

Act as a senior software reviewer. Review the code below for readability, performance, security, and maintainability. Return your answer in four sections: strengths, issues, suggested improvements, and improved code.

Code:

paste code here

Why This Is Better

Transforms a vague request into a structured review with criteria.

Prompt 5: Regex Creation

Domain: Coding

Prompt Type: Few-shot

Use Case: Creating useful regular expressions.

Bad Prompt

Make regex for emails.

Optimized Prompt

Create a regular expression based on the examples.

Positive examples: user@example.com, rajesh.a@domain.co.in

Negative examples: user@@example.com, rajesh@, domain.com

Task: Create a regex for valid basic email addresses and explain what each part does.

Why This Is Better

Few-shot examples clarify the boundary between valid and invalid outputs.

Prompt 6: Professional Email

Domain: Writing

Prompt Type: Zero-shot

Use Case: Writing workplace emails.

Bad Prompt

Write leave email.

Optimized Prompt

Write a polite professional email requesting one day of leave. Include a clear subject line, greeting, reason, date of leave, assurance about pending work, and closing. Tone: respectful and concise.

Why This Is Better

Adds tone, parts of the email, and purpose.

Prompt 7: LinkedIn Post

Domain: Writing

Prompt Type: Few-shot

Use Case: Creating professional social media posts.

Bad Prompt

Write LinkedIn post.

Optimized Prompt

Write a LinkedIn post in the style of these examples.

Example tone: professional, grateful, learning-focused, not overly dramatic.

Example structure: achievement, task details, skills learned, thanks, hashtags.

Now write a LinkedIn post about completing an Iris Flower Classification machine learning project using KNN, Decision Tree, and SVM. Include a placeholder for the GitHub link.

Why This Is Better

Gives style guidance and content requirements, making the output ready to use.

Prompt 8: Resume Bullet Improvement

Domain: Writing

Prompt Type: Reasoning summary

Use Case: Improving resume content.

Bad Prompt

Improve this resume point.

Optimized Prompt

Improve the resume bullet below using action verbs, measurable impact, and concise language. Briefly explain why the new version is stronger.

Original bullet: Worked on machine learning project.

Why This Is Better

Defines the improvement criteria and requests a short explanation.

Prompt 9: Formal Report Intro

Domain: Writing

Prompt Type: Zero-shot

Use Case: Writing project report sections.

Bad Prompt

Write introduction.

Optimized Prompt

Write a 150-word introduction for a project report on prompt engineering. Include the importance of prompts, applications in AI tools, and the goal of building a reusable prompt library. Tone: academic but simple.

Why This Is Better

Adds length, topic scope, required points, and tone.

Prompt 10: Tone Rewriting

Domain: Writing

Prompt Type: Few-shot

Use Case: Adapting tone for communication.

Bad Prompt

Make it better.

Optimized Prompt

Rewrite the message in three tones: professional, friendly, and concise.

Original message: I cannot come today because I have other work.

Return the answer in a table with columns: Tone and Rewritten Message.

Why This Is Better

Clear transformation instructions and format reduce ambiguity.

Prompt 11: Article Summary

Domain: Summarization

Prompt Type: Zero-shot

Use Case: Summarizing articles.

Bad Prompt

Summarize this.

Optimized Prompt

Summarize the text below in 5 bullet points. Then provide a 2-sentence overall summary and list 3 important keywords.

Text:
[paste text here]

Why This Is Better

Specifies length, format, and additional keyword extraction.

Prompt 12: Meeting Notes Summary

Domain: Summarization

Prompt Type: Few-shot

Use Case: Creating meeting minutes.

Bad Prompt

Summarize meeting notes.

Optimized Prompt

Convert meeting notes into structured minutes. Use this format:

Title:
Date:
Attendees:
Key Discussion Points:
Decisions Made:
Action Items with Owner and Deadline:

Notes:
[paste notes here]

Why This Is Better

Provides a reusable format for practical meeting documentation.

Prompt 13: Research Paper Summary

Domain: Summarization

Prompt Type: Reasoning summary

Use Case: Understanding research papers.

Bad Prompt

Explain this paper.

Optimized Prompt

Summarize the research paper text for a beginner. Include: problem statement, method, dataset/tools, key findings, limitations, and practical applications. Keep technical terms simple and briefly explain necessary terms.

Why This Is Better

Breaks a complex task into focused sections and defines the target audience.

Prompt 14: Long-to-Short Summary

Domain: Summarization

Prompt Type: Zero-shot

Use Case: Shortening long content.

Bad Prompt

Make this shorter.

Optimized Prompt

Condense the text below to under 100 words while preserving the main message, important facts, and original meaning. Avoid adding new information.

Text:
[paste text here]

Why This Is Better

Adds word limit and prevents hallucinated additions.

Prompt 15: Book/Chapter Summary

Domain: Summarization

Prompt Type: Few-shot

Use Case: Academic study notes.

Bad Prompt

Summarize chapter.

Optimized Prompt

Summarize the chapter using this structure:

1. Main idea
2. Key events or arguments
3. Important terms
4. Lessons learned
5. 5-question quiz with answers

Chapter text:
[paste text here]

Why This Is Better

Turns summary into a study resource with comprehension checks.

Prompt 16: Data Insight Analysis

Domain: Analysis

Prompt Type: Zero-shot

Use Case: Business data interpretation.

Bad Prompt

Analyze this data.

Optimized Prompt

Analyze the table below and identify the top 5 insights. For each insight, include supporting evidence from the data and a possible business recommendation.

Table:
[paste table here]

Why This Is Better

Defines insight count, evidence requirement, and action-oriented output.

Prompt 17: SWOT Analysis

Domain: Analysis

Prompt Type: Few-shot

Use Case: Business/project evaluation.

Bad Prompt

Do SWOT.

Optimized Prompt

Create a SWOT analysis for the given business idea. Use four sections: Strengths, Weaknesses, Opportunities, and Threats. Add 4 points under each section.

Business idea: An AI-powered study planner app for college students.

Why This Is Better

Specifies framework and number of points, improving completeness.

Prompt 18: Decision Matrix

Domain: Analysis

Prompt Type: Reasoning summary

Use Case: Tool comparison and decision-making.

Bad Prompt

Which option is better?

Optimized Prompt

Compare the following options using a decision matrix. Criteria: cost, ease of use, scalability, learning value, and risk. Score each criterion from 1 to 5 and briefly justify each score. End with a recommendation.

Options: ChatGPT, Gemini, Claude

Why This Is Better

Uses a structured evaluation method and asks for evidence-based recommendation.

Prompt 19: Root Cause Analysis

Domain: Analysis

Prompt Type: Zero-shot

Use Case: Troubleshooting technical problems.

Bad Prompt

Why did this fail?

Optimized Prompt

Perform a root cause analysis for the issue below. Use the 5 Whys method, list the likely root cause, and suggest 3 preventive actions.

Issue: A machine learning model performs well on training data but poorly on testing data.

Why This Is Better

Specifies an analytical method and desired solution output.

Prompt 20: Pros and Cons

Domain: Analysis

Prompt Type: Few-shot

Use Case: Evaluating educational technology.

Bad Prompt

Tell pros cons.

Optimized Prompt

Analyze the pros and cons of using AI tools for student learning. Return the answer as a table with columns: Aspect, Pros, Cons, and Balanced Recommendation. Include at least 5 aspects.

Why This Is Better

Changes a broad opinion request into a balanced structured analysis.

Prompt 21: Story Idea

Domain: Creative

Prompt Type: Zero-shot

Use Case: Generating story concepts.

Bad Prompt

Give story.

Optimized Prompt

Create a short story idea about a student who builds an AI assistant. Include genre, main character, conflict, twist, and ending. Keep it original and suitable for a general audience.

Why This Is Better

Specifies creative elements and audience suitability.

Prompt 22: Poem Generation

Domain: Creative

Prompt Type: Few-shot

Use Case: Creative writing with constraints.

Bad Prompt

Write poem.

Optimized Prompt

Write a motivational poem for students. Style requirements: simple words, 4 stanzas, 4 lines per stanza, hopeful tone, no complex vocabulary. Theme: learning from failure.

Why This Is Better

Adds format, tone, audience, and theme.

Prompt 23: Brand Name Ideas

Domain: Creative

Prompt Type: Reasoning summary

Use Case: Naming a product or project.

Bad Prompt

Suggest names.

Optimized Prompt

Generate 15 brand name ideas for an AI learning app. For each name, include a one-line meaning and why it fits the product. Keep names short, memorable, and easy to pronounce.

Why This Is Better

Adds quantity, evaluation criteria, and explanation.

Prompt 24: YouTube Script

Domain: Creative

Prompt Type: Zero-shot

Use Case: Short-form educational content.

Bad Prompt

Make video script.

Optimized Prompt

Write a 60-second YouTube Shorts script explaining prompt engineering. Include hook, simple explanation, example, benefit, and call to action. Use a friendly student tone.

Why This Is Better

Defines platform, duration, structure, and tone.

Prompt 25: Ad Copy

Domain: Creative

Prompt Type: Few-shot

Use Case: Marketing copy generation.

Bad Prompt

Write ad.

Optimized Prompt

Create 5 advertisement captions for an AI note-taking app. Each caption should be under 15 words, include a clear benefit, and sound modern and student-friendly.

Why This Is Better

Adds count, length limit, benefit focus, and audience.

Reusable Prompt Template Library

1. General Task Template

Act as a [role]. Your task is to [specific task].
Context: [background information].
Requirements:

- [requirement 1]
- [requirement 2]
- [requirement 3]

Output format: [table/bullets/paragraph/code].
Tone: [formal/friendly/academic/simple].
Constraints: [word limit/style/tools].

2. Coding Prompt Template

Act as a [programming language] expert.
Task: [debug/explain/optimize/write] the following code.
Requirements:

- Explain the issue briefly.
- Provide corrected or improved code.
- Mention best practices.

Code:
[paste code here]

3. Writing Prompt Template

Write a [content type] about [topic].
Audience: [target audience].
Tone: [tone].
Length: [word count].
Must include:

- [point 1]
- [point 2]
- [point 3]

Avoid: [things to avoid].

4. Summarization Prompt Template

Summarize the following text for [audience].
Format:

- 5 bullet points
- 2-sentence summary
- 3 keywords

Rules:

- Keep the original meaning.
- Do not add new information.

Text:
[paste text here]

5. Analysis Prompt Template

Analyze [topic/data/problem] using [framework].
Include:

- Key observations
- Evidence
- Risks

- Recommendations
Output format: [table/numbered list/report].

6. Creative Prompt Template

Create a [story/poem/script/idea list] about [topic].
Style: [style].
Audience: [audience].
Constraints:
- [length]
- [tone]
- [format]
Include: [required elements].

Conclusion

Effective prompt engineering improves AI output by making instructions clear, specific, contextual, and structured. Strong prompts define the role, task, audience, constraints, output format, and evaluation criteria. This prompt library can be reused and adapted for coding, writing, summarization, analysis, and creative projects.